5.1 Limit Aljabar

a) Hit. Limit Manual

L_1

limit :

$$L_1 = \lim_{x \to k} \frac{x^2 - k^2}{x - k}$$

dengan :

$$k = 4$$

Faktorkan :

$$x^2 - k^2 = (x-k)(x+k)$$

Substitusi :

$$\lim_{x \to 4} \frac{(x-4)(x+4)}{x-4}$$

$$= \lim_{x \to 4} (x+4)$$

Subs : $4+4$

$L_1 = 8$

L_2 :

$$L_2 = \lim_{x \to 0} \frac{\sqrt{x + (k+1)^2 - (k+1)}}{x}$$

Dengan :

$$k = 4$$

$$(k+1) = 5$$

kalikan sekawan :

$$= \frac{\sqrt{x+25} - 5}{x} \times \frac{\sqrt{x+25} + 5}{\sqrt{x+25} + 5}$$

atas :

$$= (x+25) - 25 = x$$

Bawah :

$$= x(\sqrt{x+25} + 5)$$

Sederhanakan :

$$= \frac{1}{\sqrt{x+25} + 5}$$

Subs :

$$= \frac{1}{5+5}$$

$L_2 = 1/10$

L_3 :

$$L_3 = \lim_{x \to \infty} \frac{(a+2)x^2 + bx + 7}{x^3 - x + k}$$

Subs :

$a = 0$
$b = 1$
$k = 4$

$$= \lim_{x \to \infty} \frac{2x^2 + x + 7}{x^3 - x + 4}$$

Pangkat tertinggi :

$$= \frac{2}{1}$$

$L_3 = 2$

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5.2 Limit Trigono

L4:

$$L4 = \lim_{x \to 0} \frac{\sin((a+1)x)\tan((b+2)x)}{x^2}$$

subs:

$a = 0 \rightarrow (a+1) = 1$
$b = 1 \rightarrow (b+2) = 3$

$$= \lim_{x \to} \frac{\sin(x)\tan(3x)}{x^2}$$

Gunakan:

$\sin x = x$
$\tan x = x$

$$= \frac{x(3x)}{x^2}$$

$L4 = 3$

5.3 Limit tak hingga

d) Limit CV

$$R(n) = \frac{100n + k}{n + (a + b + 1)}$$

subs

$k = 4$
$a = 0$
$b = 1$

$(a + b + 1) = 2$

$$= \frac{100n + 4}{n + 2}$$

pangkat tinggi

$= 100$

Limit $= 100$

Interpretasi

Pendapatan Stabil =

100 ribu perbulan.

e) Limit akar : $\lim_{n \to \infty} (\sqrt{n^2 + kn} - \sqrt{n^2 + bn})$

subs : $k = 4$
$b = 1$

Hasil akhir $= 3/2$

Selisih model CLV stabil di 1,5

5.4 kekontinuan

Fungsi

$$C(x) = \frac{x^2 - k^2}{x - k}, \; x < k$$

Saat limit :

$$\lim_{x \to k} (x + k)$$

subs : $= 2k$
$= 8$

Didapat : $P = 2k = 8$

Untuk :

$ax + q$ pada $x > k$

Harus :

$ak + q = 2k$

Subs :

$a = 0$

$q = 8$

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